Linear Algebra — Final-Exam Problems with Detailed Solutions (Integrated Version)

Problem 1: Determinant Computation

Problem: Compute the n-order determinant:

$$D_n = \begin{vmatrix} a_1+\lambda & a_2 & \cdots & a_n \\ a_1 & a_2+\lambda & \cdots & a_n \\ \vdots & \vdots & \ddots & \vdots \\ a_1 & a_2 & \cdots & a_n+\lambda \end{vmatrix}$$

Solution:

1. **Column operation**: Add columns $2, 3, \ldots, n$ all to column 1; every entry of column 1 becomes $\lambda + \sum_{i=1}^{n} a_i$.
2. **Factor out the common factor**:

$$D_n = (\lambda + \sum_{i=1}^{n} a_i) \begin{vmatrix} 1 & a_2 & \ldots & a_n \\ 1 & a_2+\lambda & \ldots & a_n \\ \vdots & \vdots & \ddots & \vdots \\ 1 & a_2 & \ldots & a_n+\lambda \end{vmatrix}$$

3. **Row operations**: subtract row 1 from row i $(i = 2, \ldots, n)$, reducing the determinant to upper triangular form.
4. **Result**: multiply the diagonal entries:

$$D_n = \lambda^{n-1}(\lambda + a_1 + a_2 + \cdots + a_n)$$

Problem 2: Vector Operations

Problem: Let $\alpha = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$; compute $\alpha^T\alpha, \alpha\alpha^T, \alpha\alpha^T\alpha, (\alpha\alpha^T)^{100}$.

Solution:

1. **Scalar product**: $\alpha^T\alpha = 1^2 + 2^2 + 1^2 = 6$.
2. **Matrix product**: $\alpha\alpha^T = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix}$.
3. **Mixed product**: by associativity, $\alpha\alpha^T\alpha = \alpha(\alpha^T\alpha) = 6\alpha = \begin{pmatrix} 6 \\ 12 \\ 6 \end{pmatrix}$.
4. **Power**: let $B = \alpha\alpha^T$; then $B^2 = 6B$, and by induction $B^n = 6^{n-1}B$.

$$(\alpha\alpha^T)^{100} = 6^{99} \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

Problem 3: Proving Matrix Invertibility

Problem: Let A be an n-order square matrix satisfying $A^2 - 5A + E = 0$; prove that $A - E$ is invertible and find its inverse.

Solution:

1. Rearrange the equation: $A^2 - 5A + E = 0 \Rightarrow A^2 - A - 4A + 4E = 3E$.
2. Factor: $(A - E)(A - 4E) = 3E$.
3. Hence: $(A - E) \cdot \frac{1}{3}(A - 4E) = E$.
4. By definition, $A - E$ is invertible and $(A - E)^{-1} = \frac{1}{3}(A - 4E)$.

Problem 4: Discussion of a Linear System

Problem: Discuss the solutions of the system $\begin{cases} ax_1 + x_2 + x_3 = 0 \\ x_1 + ax_2 + x_3 = 2 \\ x_1 + x_2 + ax_3 = a \end{cases}$.

Solution: the determinant of the coefficient matrix is $|A| = (a + 2)(a - 1)^2$.

1. **Unique solution**: when $a \neq 1$ and $a \neq -2$, the system has a unique solution.
2. **No solution**: when $a = 1$, row reduction of the augmented matrix gives $r(A) = 1 \neq r(\overline{A}) = 2$, so there is no solution.
3. **Infinitely many solutions**: when $a = -2$, the equivalent system is $\begin{cases} x_1 - 2x_2 + x_3 = 2 \\ -3x_2 + 3x_3 = 4 \end{cases}$; taking the free variable $x_3 = c$, the general solution is $X = c \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} -2/3 \\ -4/3 \\ 0 \end{pmatrix}$.

Problem 5: Proof of Linear Independence

Problem: Given that $\alpha_1, \alpha_2, \alpha_3$ are linearly independent, and $\beta_1 = \alpha_1 + \alpha_2, \beta_2 = \alpha_2 + \alpha_3, \beta_3 = \alpha_3 + \alpha_1$, prove that $\beta_1, \beta_2, \beta_3$ are linearly independent.

Solution:

1. Set $k_1\beta_1 + k_2\beta_2 + k_3\beta_3 = 0$; substituting and collecting terms gives $(k_1 + k_3)\alpha_1 + (k_1 + k_2)\alpha_2 + (k_2 + k_3)\alpha_3 = 0$.
2. Since the α_i are linearly independent, the determinant of the coefficient matrix is $\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 2 \neq 0$.
3. The system has only the zero solution, so the β_i are linearly independent.

Problem 6: Determining Parameters and the Maximal Linearly Independent Subset

Problem: $A = \begin{pmatrix} 1 & 1 & 1 & 1 & \\ 3 & 2 & 1 & -3 & a \\ 0 & 1 & 2 & 6 & 3 \\ 5 & 4 & 3 & -1 & b \end{pmatrix}$ has rank 2; find a, b and a maximal linearly independent subset of the columns.

Solution:

1. **Row operations**: row reduction to echelon form shows rows 3 and 4 must be zero rows.
2. **Parameters**: solving gives $a = 0, b = 2$.
3. **Maximal linearly independent subset**: the pivots lie in columns 1 and 2, so $\{\alpha_1, \alpha_2\}$ is a maximal linearly independent subset.
4. **Linear expressions**: the remaining columns are $\alpha_3 = -\alpha_1 + 2\alpha_2, \alpha_4 = -5\alpha_1 + 6\alpha_2, \alpha_5 = -2\alpha_1 + 3\alpha_2$.

Problem 7: Eigenvalues and Eigenvectors

Problem: Find the eigenvalues and eigenvectors of $A_1 = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and $A_2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$.

Solution:

1. A_1: the eigenvalues are $\lambda_1 = 0, \lambda_2 = 1, \lambda_3 = 2$.
   - $\lambda_1 = 0 \to \xi_1 = (-1, 1, 0)^T$
   - $\lambda_2 = 1 \to \xi_2 = (0, 0, 1)^T$
   - $\lambda_3 = 2 \to \xi_3 = (1, 1, 0)^T$
2. A_2: the eigenvalues are $\lambda_1 = 0, \lambda_2 = 2$.
   - $\lambda_1 = 0 \to \xi_1 = (-1, 1)^T$
   - $\lambda_2 = 2 \to \xi_2 = (1, 1)^T$